

How to Change Structure of Live DB (aka Database Migration)

How to change schemas?

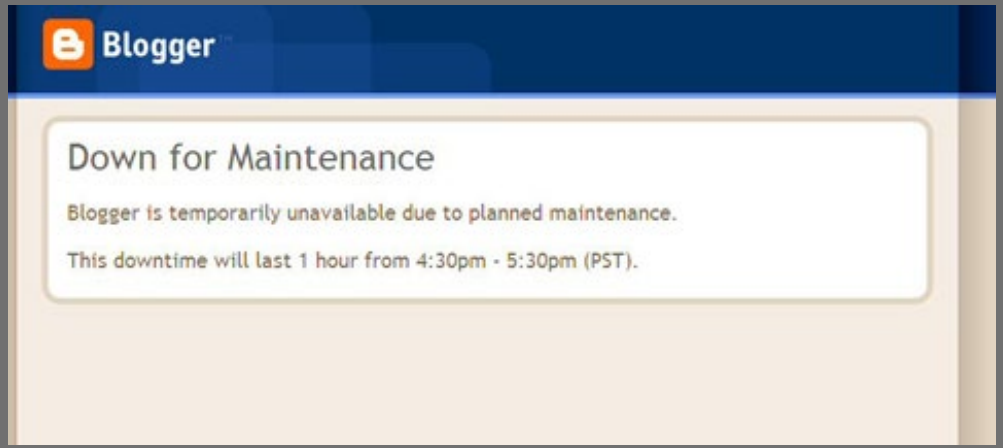
- Create new tables.
- Create new queries using the new tables.
- Copy or compute data for the new tables.
- Switch to using new tables and queries.

} Can take substantial time. Lots of data. Possibly distributed & replicated.

What about data that arrives during this process?

Traditional – shutdown

Annoying!



Better – locking

Only shutdown what interferes with the changes.

Lock writes to each changing table.

Best – invisible to user

A simple version:

1. Create shadow tables – tables with new schema.
2. Create triggers – old tables forwards data updates to new tables.
3. Copy old data to new tables.
4. Rename tables so that new tables are now used.

Can choose to only let some users see the new version (A/B testing).

DBMS might provide support for such online schema changes.